Category

Arbitrage, Replication, and the Cost of Carry in Pricing Derivatives

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ARBITRAGE

Learning Outcome

- explain how the concepts of arbitrage and replication are used in pricing derivatives

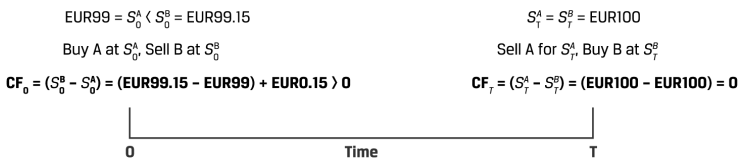
An earlier lesson on market efficiency established that market prices should not allow for the possibility of riskless profit or arbitrage in the absence of transaction costs. In its simplest form, an arbitrage opportunity arises if the "law of one price" does not hold, or an identical asset trades at the same time at different prices in different places.

In the case of a derivative contract whose value is derived from future cash flows associated with the price of an underlying asset, arbitrage opportunities arise either if two assets with identical *future cash flows* trade at different prices or if an asset with a known future price does not trade at the *present value* of its future price determined using an appropriate discount rate.

The first case of assets with identical future cash flows trading at different prices is illustrated in Exhibit 1.

Exhibit 1:

Assets with Identical Future Cash Flows Trade at Different Prices



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For example, assume the two assets are zero-coupon bonds with identical features and the same issuer. Both bonds mature on the same future date with a payoff of par and have the same risk of default between now and the maturity date.

1. Bond A has a price of EUR99 at time $t = 0$ (S_0^A = EUR99).
2. Bond B has a price of EUR99.15 at time $t = 0$ (S_0^B = EUR99.15).
3. Both bonds have an expected future price of EUR100 ($S_T^A = S_T^B$ = EUR100).

This scenario represents an arbitrage opportunity for an investor.

- At time $t = 0$, the investor can:
  - Sell Bond B short to receive proceeds of EUR99.15 and purchase Bond A for EUR99;
  - Realize a net cash inflow at $t = 0$ of EUR0.15.
- At time $t = T$, when both bonds mature, the investor:
  - Receives EUR100 for Bond A and uses this to buy Bond B for EUR100 to cover the short position.
  - The offsetting cash flows at time T leave the investor with a riskless profit of the EUR0.15 price difference between Bonds A and B at time 0.

same price ($S_0^{-} = S_0^{+}$).

A second type of derivative-related arbitrage opportunity arises when an asset with a known future price does not trade at the present value (PV) of its future price. An earlier time-value-of-money lesson distinguished between discrete and continuous compounding in calculating present versus future value. The future value of a single cash flow based on a *discrete* number of uniform periods follows the general formula in Equation 1:

$$FV_N = PV(1 + r)^N, \qquad 1$$

where r is the stated interest rate per *period* and N is the number of compounding periods. *Continuous compounding* is the case in which the length of the uniform periods approaches zero, so the number of periods per year approaches infinity and is calculated using the natural logarithm, as shown in Equation 2:

$$FV_T = PVe^{rT}. \qquad 2$$

While derivatives may be priced with either approach, for purposes of this and later lessons, we will use the discrete compounding method for individual underlying assets. However, for underlying assets that represent a portfolio, such as an equity, fixed-income, commodity, or credit index, or where the underlying involves foreign exchange where interest rates are denominated in two currencies, continuous compounding will be the preferred method. Ignoring additional costs or benefits associated with asset ownership, the appropriate discount rate, r, is the risk-free rate, as demonstrated in Example 1.

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EXAMPLE 1

In an earlier lesson, Procam Investments entered into a contract to buy 100 ounces of gold at an agreed-upon price of USD1,792.13 per ounce in three months. In this example, Procam does the opposite trade, given a discrepancy between spot and discounted known future gold prices. Assume that today's spot gold price (S_0) is USD1,770 per ounce, and the annualized risk-free interest rate (r) is 2%. For purposes of this example, we assume that Procam can borrow at the risk-free rate and gold may be stored at no cost. Under these conditions, we demonstrate how Procam can generate a riskless profit:

- At time $t = 0$:
- Procam borrows USD177,000 at 2.0% interest for three months and purchases 100 ounces of gold at today's spot price.
- Procam enters into a forward contract today to sell 100 ounces of gold at a price of USD1,792.13 per ounce in three months.

$S_0 < S_T(1 + r)^{-T}$	$S_0(1 + r)^T < S_T$
Borrow S_0 at r and Buy S_0, Agree to sell at S_T	Sell at S_T and Repay $S_0(1 + r)^T$
Borrow USD1,770 at 2% and Buy gold (S_0)	Sell at USD1,792.13 (S_T) and Repay $S_0(1 + r)^T$ = USD1,778.78
Contract to sell at S_T = USD1,792.13 per ounce	Riskless profit of USD13.35 per ounce
$CF_0 = 0$	**CF_T = (USD1,792.13 − USD1,778.78) > 0**

0 — Time — T

- At time $t = T$ (in three months):
- Procam delivers 100 ounces of gold under the forward contract and receives USD179,213 (= 100 × USD1,792.13).
- Procam repays the loan principal with interest:

$$\text{USD}177{,}878.44 = \text{USD}177{,}000(1.02)^{0.25}.$$

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In Example 1, the spot price of gold, S_0, is below the present value of the known future price of gold in three months' time ($S_0 < S_T(1 + r)^{-T}$, since USD1,770 < USD1,783.28). Procam earns a riskless profit of USD13.35 per ounce by borrowing and purchasing gold in the cash market and simultaneously selling gold in the forward market at an agreed-upon price. We would expect that as other investors recognize and pursue this opportunity, the spot price will increase (and the forward price will fall) until the spot price is equal to the discounted value of the known future price ($S_0 = S_T(1 + r)^{-T}$) to eliminate this arbitrage opportunity.

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The two key arbitrage concepts used to price derivatives for an underlying with no additional cash flows may be summarized as follows:

- Identical assets or assets with identical cash flows traded at the same time must have the same price ($S_0^A = S_0^B$).
- Assets with a known future price must have a spot price that equals the future price discounted at the risk-free rate ($S_0 = S_T(1 + r)^{-T}$).

These arbitrage conditions establish the relationship between spot prices, forward commitment prices, and the risk-free rate shown in Exhibit 2.

Exhibit 2:

Spot Prices, Forward Commitment Prices, and the Risk-Free Rate

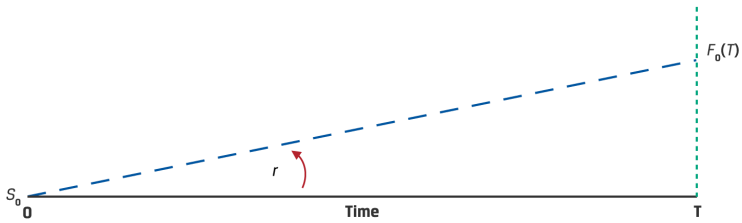

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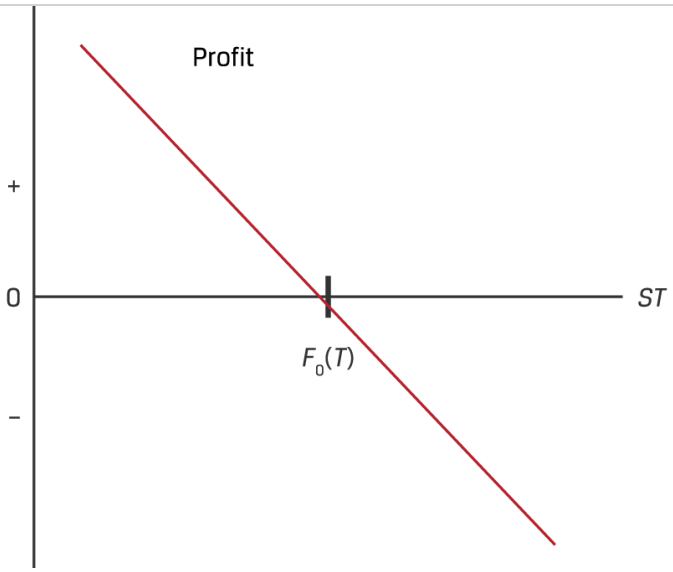
Exhibit 2 shows the case of $r > 0$ for an asset with no additional income or costs under discrete compounding. Note that for a given time T, the forward price will be higher relative to the spot price with a higher risk-free rate r. Also, for a given risk-free rate r, as T increases, the forward price will increase relative to the spot price. It is important to note that the relevant risk-free rate for most market participants is the repo rate, introduced in an earlier lesson, where borrowed funds are collateralized by highly liquid securities.

Recall from an earlier lesson that a forward commitment has a symmetric payoff profile. That is, at time T, the transaction is settled on the basis of the difference between the forward price $F_0(T)$ and the underlying price of S_T, or $F_0(T) - S_T$ from the seller's perspective, as in Exhibit 3.

Exhibit 3:

Forward Commitment Seller Payoff Profile



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The forward seller realizes a gain if the seller is able to deliver the underlying at a market value (S_T) below the pre-agreed price, $F_0(T)$. In Example 1, Procam borrows to buy the underlying at a spot price below the present value of S_T to lock in a riskless gain. So far, we have taken the forward price, $F_0(T)$, as given. In what follows, we demonstrate how no-arbitrage conditions may be used to establish the relationship between spot and forward prices by replicating or recreating an exact offsetting position for a forward commitment.

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